Median Household Income and Primary Drinking Water Violations in California

A simple linear regression analysis of 2026 SAFER data

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Abstract

This paper examines whether service-area median household income (MHI) differs between California community water systems with and without a current primary maximum contaminant level (MCL) violation. Using the September 22, 2026 SAFER public dataset, I fit a simple linear regression with MHI as the response and violation status as a binary predictor. Among 2,156 systems with usable MHI data, systems with a primary MCL violation had an estimated mean MHI about 25,456 dollars lower than systems without one (95% CI: 16,209 to 34,704 dollars lower; $p < 0.001$). The results show a clear association, although the analysis is observational and violation status explains only a small portion of the overall variation in MHI.

1 Introduction

California public drinking water systems serve communities with different socioeconomic conditions and different levels of water-quality compliance. A maximum contaminant level (MCL) is a regulatory limit on how much of a contaminant can be present in drinking water. Under California's Failing Water System criteria, a primary MCL violation with an open enforcement action is one condition that can place a system on the Failing list (California State Water Resources Control Board, 2024). Since primary MCLs are health-based standards, it is important to understand whether these violations are more common in communities with different income levels.

Previous research has found relationships between drinking water compliance and socioeconomic conditions. Allaire, Wu, and Lall (2018) found that health-based violations were associated with several community and system characteristics nationally. In California's San Joaquin Valley, Balazs et al. (2012) found that arsenic violations were more common among systems serving disadvantaged populations. The State Water Board's 2026 Needs Assessment similarly re-

ported that Failing and At-Risk systems are more likely to serve disadvantaged communities (California State Water Resources Control Board, 2026a).

This paper asks: **How does service-area median household income differ between California community water systems with and without a current primary MCL violation?** I use simple linear regression with MHI as the response and violation status as a binary predictor. The results show that systems with a primary MCL violation serve areas with substantially lower estimated MHI on average, although violation status explains only a small portion of the total variation in MHI.

2 Data

The data came from the California State Water Resources Control Board's **Drinking Water - SAFER Dashboard Failing and At-Risk Drinking Water Systems** dataset (California State Water Resources Control Board, 2026b). The file contains 3,190 water systems and 140 variables. For this analysis, I only used community water systems because these systems provide water to year-round residential communities. This left 2,802 community water systems in the dataset.

The main response variable was **median household income (MHI)**. The State Water Board estimates MHI using American Community Survey data along with water-system service areas, residential parcel information, and accepted income surveys when available (California State Water Resources Control Board, 2026c). Since MHI is estimated for the service area rather than measured directly from every household, the values may not perfectly represent every household served by the system.

The explanatory variable was **primary MCL violation status**, taken from `PRIMARY_MCL_VIOLATION`. A value of `YES` means that the water system currently meets the primary MCL violation criterion used for the Failing list, while `NO` means that it does not currently meet this criterion (California State Water Resources Control Board, 2024). For the regression model, I coded this variable as

$$X_i = \begin{cases} 1, & \text{if system } i \text{ has a primary MCL violation,} \\ 0, & \text{otherwise.} \end{cases}$$

The indicator does not describe the contaminant concentration, severity, or duration of the violation.

The main data-cleaning issue was missing MHI. Of the 2,802 community water systems, 646 (23.1%) had missing or indeterminate MHI, leaving 2,156 systems for analysis. Missingness was slightly higher among systems with a primary MCL violation (25.1%) than among systems without one (22.9%). I did not

Table 1: Median household income by primary MCL violation status.

Primary MCL status	n	Mean MHI	Median MHI	SD
No primary MCL violation	1992	$88,018	$77,981	$58,651
Primary MCL violation	164	$62,561	$47,683	$50,068

replace these missing values because there was not enough information in the dataset to do so without making additional assumptions. Other information such as system size, ownership, source-water type, and geographic region could be included in a more detailed analysis.

The final sample contains 1,992 systems without a primary MCL violation and 164 systems with one. Mean MHI is $88,018 for systems without a violation and $62,561 for systems with a violation. The medians show the same pattern: $77,981 compared with $47,683.

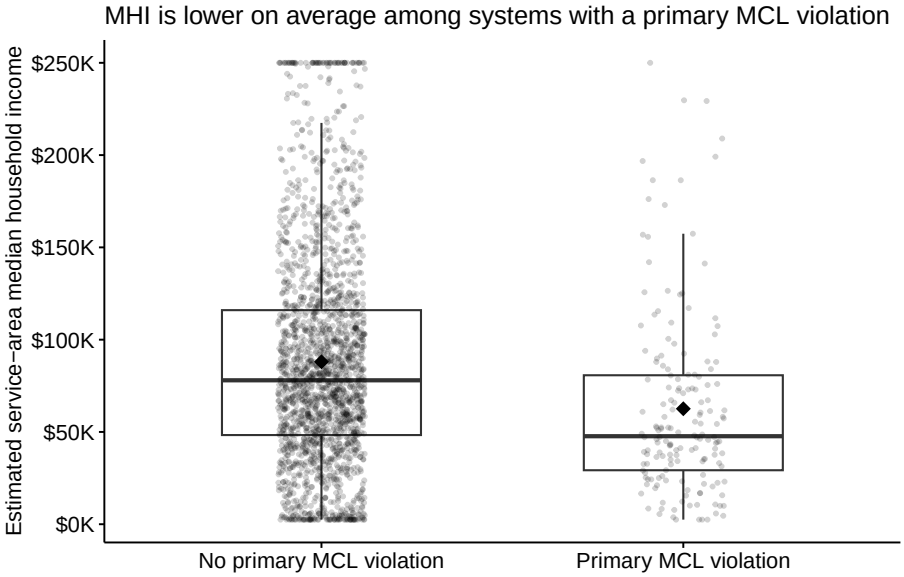


Figure 1: Estimated service-area MHI by primary MCL violation status. Boxes show the median and interquartile range; points are individual systems, and diamonds show group means.

Figure 1 shows that MHI is generally lower for systems with a primary MCL violation, although there is substantial variation within both groups. This suggests that violation status is related to average MHI but does not explain income differences by itself.

3 Methods

The main statistical analysis was a simple linear regression model comparing median household income with primary MCL violation status. MHI was used as the response variable, while primary MCL violation status was used as the explanatory variable. MHI was divided by 10,000 before modeling so that the regression coefficients would be easier to read. This model was appropriate because the research question asks whether average household income differs between systems with and without a primary MCL violation.

Let Y_i represent the estimated MHI, in units of 10,000 dollars, for water system i, and let X_i represent its primary MCL violation status. The model is

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i.$$

In this model, β_0 represents the average MHI for systems without a primary MCL violation. The coefficient β_1 represents the difference in average MHI between systems with and without a violation. Since violation status is coded as 0 or 1, a negative value of β_1 means that systems with a primary MCL violation have a lower average MHI.

To test whether the difference between the two groups was statistically significant, I used the following hypotheses:

$$H_0 : \beta_1 = 0 \qquad \text{versus} \qquad H_A : \beta_1 \neq 0.$$

The test used a significance level of $\alpha = 0.05$. I also calculated a 95% confidence interval and R^2 to evaluate the size of the difference and how much of the variation in MHI is explained by violation status.

The regression assumptions were also checked before interpreting the results. Since the predictor only has two groups, the linearity assumption is less of a concern than it would be with a continuous predictor. However, the groups do not have exactly the same variation in MHI, and the residuals are somewhat right-skewed. Independence may also not be perfect because nearby water systems can have similar economic and geographic conditions. In addition, some community systems have missing MHI values, which may affect how representative the final sample is.

No model selection was needed because the analysis only uses one explanatory variable based directly on the research question. I checked the model using residual plots and a sensitivity analysis that removed MHI values recorded at the ACS boundaries of \$2,500 and \$250,000. All data cleaning, analysis, and figures were completed in R using the `tidyverse`, `ggpubr`, and `gridExtra` packages.

Table 2: Simple linear regression results for MHI and primary MCL violation status.

Term	Estimate	SE	95% CI	p
No violation	8.802	0.130	[8.547, 9.057]	< 0.001
Violation	-2.546	0.472	[-3.47, -1.621]	< 0.001

4 Results

The fitted regression is

$$\widehat{Y}_i = 8.802 - 2.546 X_i.$$

The intercept represents an estimated mean MHI of \$88,018 for systems without a primary MCL violation. The slope shows that systems with a primary MCL violation have an estimated mean MHI **\$25,456 lower** than systems without one.

The 95% confidence interval for the difference is from -\$34,704 to -\$16,209. The test statistic is $t = -5.4$ with 2154 degrees of freedom, and the p-value is below 0.001. At $\alpha = 0.05$, I reject H_0. This shows that primary MCL violation status is significantly related to service-area MHI in this dataset.

However, the model has an R^2 of only 0.013, meaning that violation status explains about 1.3% of the observed variation in MHI. This shows that the two groups have different average MHI values, but violation status alone does not explain most of the income differences between water systems.

The diagnostic plots show some differences in variance between the groups and some deviation from normality in the upper tail. The sample standard deviations are \$58,651 for the no-violation group and \$50,068 for the violation group. This means the regression assumptions are not perfectly satisfied, so the results should be interpreted with some caution.

As a sensitivity check, I removed the 122 systems recorded at exactly \$2,500 or \$250,000 and fit the model again. The estimated difference remained \$22,673 lower for systems with a primary MCL violation, with a 95% confidence interval from -\$30,886 to -\$14,459 and a p-value below 0.001. This suggests that the extreme boundary values are not driving the main result.

5 Discussion

This analysis found that California community water systems with a current primary MCL violation serve areas with lower estimated median household income on average. In the September 22, 2026 SAFER dataset, the estimated difference

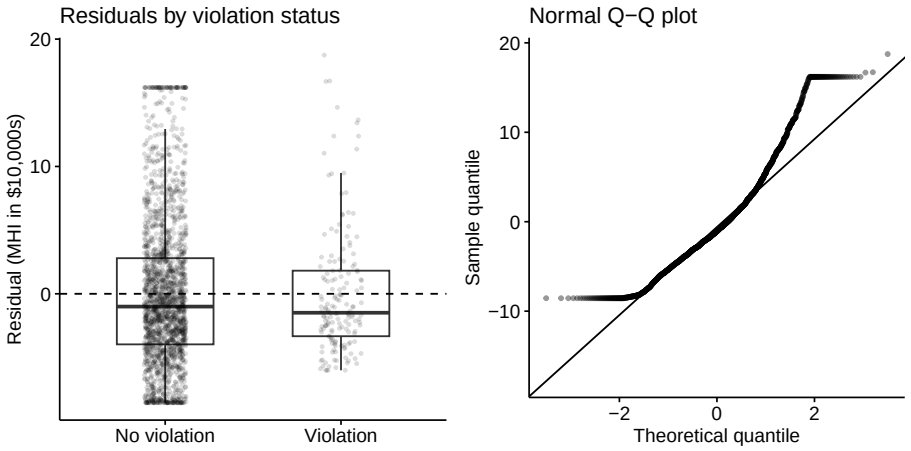


Figure 2: Regression diagnostics showing residuals by violation status and a normal Q-Q plot.

was approximately $25,456. The median comparison and sensitivity analysis showed the same overall pattern. These results are consistent with previous research connecting drinking-water compliance problems with socioeconomic disadvantage (Balazs et al., 2012; California State Water Resources Control Board, 2026a).

There are several limitations to this analysis. The results do not show that lower household income directly causes MCL violations because the data are observational. The model also does not include other factors such as system size, ownership, source-water type, geographic region, infrastructure, or contaminant type. Missing MHI values are another limitation, and MHI itself is estimated using ACS and geographic data rather than directly measured from every household. The violation variable also does not show the severity or duration of a violation.

The model explains only a small portion of the overall variation in MHI. In the future, the analysis could be improved by using multiple regression to include factors such as system size, ownership, and region. Another option would be to model violation status as the response or analyze different contaminants separately. These additions could help explain more of the differences between water systems.

6 References

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